

Edward P. Chandler

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edchandler00.github.io

Research Area

Computational Imaging, Deep Learning, Optimization, Inverse Problems, and Computer Vision

Education

University of Wisconsin–Madison

Ph.D. in Electrical and Computer Engineering

Dec. 2027

GPA: 4.00 / 4.00

- **Advisor:** Prof. Ulugbek S. Kamilov

Washington University in St. Louis

Transferred with Advisor after 2 years of Ph.D. research

- **Advisor:** Prof. Ulugbek S. Kamilov

Washington University in St. Louis

B.S. in Computer Science + Mathematics

May 2023

GPA: 4.00 / 4.00

- **Minor:** Philosophy-Neuroscience-Psychology
- **Research Advisor:** Prof. Ulugbek S. Kamilov
- **Academic Advisor:** Prof. Ron Cytron

Awards and Honors

NSF GRFP Honorable Mention

2024

McKelvey School of Engineering Valedictorian

2023

McKelvey School of Engineering Award for Research Excellence – CSE

2023

McKelvey Dean's Select Ph.D. Fellowship

2022

Experience

Computational Imaging Group, UW-Madison

Madison, WI

Graduate Researcher

Jan. 2026 – Present

- Design, analyze, and implement generative models (e.g. diffusion and flow-matching) to serve as data-driven priors for imaging inverse problems.
- Develop convex and non-convex optimization algorithms to solve high-dimensional inverse problems.
- Apply these frameworks to a variety of imaging modalities, particularly CT and MRI.
- Collaborate with other research labs on image reconstruction problems.
- Supervise undergraduate and Master students' research projects.

Los Alamos National Laboratory

Los Alamos, NM

Graduate Researcher

Jun. 2025 – Aug. 2025, May 2026 – Aug. 2026

- Design and implement large-scale Computed Tomography/Laminography (CT/CL) reconstruction algorithms.
- Develop proximal-based optimization algorithms.

Complex Media Optics Lab, ENS Paris

Paris, France

Visiting Graduate Researcher

Jan. 2024 – May 2024

- Implemented the system modeling and reconstruction algorithm for a novel Raman spectroscopy imaging system.
- Resulted in a publication in *Nature Communications*.

Computational Imaging Group, Washington University in St. Louis

St. Louis, MO

Undergraduate Researcher

Sep. 2020 – May 2023

Graduate Researcher

May 2023 – Dec. 2025

- Designed, analyzed, and implemented Plug-and-Play image reconstruction algorithms.

- Analyzed the effect of distribution shifts on learned image reconstruction algorithms and designed algorithms to mitigate the effect.
- Trained denoisers and an image segmentation model on real data from WashU collaborators.

General Dynamics Information Technology

St. Louis, MO

Computer Vision Data Science Intern

Sep. 2021 – Aug. 2022

- Trained and optimized deep learning segmentation models for satellite imagery datasets.

The Trade Desk

Boulder, CO

Software Engineer Intern

Jan. 2019 – Apr. 2019

- Built a full-stack news app using The Trade Desk’s database of internet users.

Teaching Experience

UW-Madison

Madison, WI

Head Teaching Assistant, ECE 888: Topics in Mathematical Data Science

Aug. 2026 – Dec. 2026

- Design assignments, give lectures, and conduct office hours.

Washington University in St. Louis

St. Louis, MO

Teaching Assistant, ESE 351: Signals and Systems

Jan. 2025 – May. 2025

- Design assignments, give tutorials, and conduct office hours.

Washington University in St. Louis

St. Louis, MO

Teaching Assistant, CSE 247: Data Structures and Algorithms

Sep. 2020 – Dec. 2020

- Led weekly group tutorials on analytical and coding problems.

Scholarly Activities

Invited Talks

- “Measurement Alignment for Computed Laminography,” New Computational Methods for Imaging at Light and Electron Sources Workshop, 2026 Bay Area Joint Light Sources Users’ Meeting

Conferences Reviewed

- IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026
- European Conference on Computer Vision (ECCV) 2026
- Conference on Neural Information Processing Systems (NeurIPS) 2026

Journals Reviewed

- IEEE Transactions on Computational Imaging (TCI)
- IEEE Signal Processing Letters (SPL)
- SIAM Journal on Imaging Sciences (SIIMS)

Publications (*Equal contribution/Shared first authorship)

- [1] S. Shoushtari, **E. P. Chandler**, and U. S. Kamilov. “Moments Matter: Posterior Recovery in Poisson Denoising via Log-Networks”. In: *IEEE ICASSP*. 2026.
- [2] A. Scott, E. Louwagie, **E. P. Chandler**, S. Kolluru, C. Fosher, A. Arter, F. Manyx, U. S. Kamilov, A. Odibo, and M. Oyen. “Predicting Stillbirth Risk Factors with Ultrasound Texture Analysis and Doppler Imaging of the Placenta”. In: *Reproductive Sciences*, 2026.
- [3] **E. P. Chandler**, S. Shoushtari, B. Wohlberg, and U. S. Kamilov. “Closed-Form Approximation of the Total Variation Proximal Operator”. In: *IEEE TCI*, 2025.
- [4] S. Zhao, L. Chibani, **E. P. Chandler**, F. Liu, J. Hu, L. Valzania, U. S. Kamilov, and H. B. de Aguiar. “Computational field-resolved coherent chemical imaging”. In: *Nature Communications*, 2025.
- [5] W. Wang*, S. Shoushtari*, **E. P. Chandler**, M. S. Asif, and U. S. Kamilov. “Distribution Shift Estimation in Imaging Inverse Problems”. In: *IEEE CAMSAP*. 2025.
- [6] S. Shoushtari, J. Liu, **E. P. Chandler**, M. S. Asif, and U. S. Kamilov. “Prior Mismatch and Adaptation in PnP-ADMM with a Nonconvex Convergence Analysis”. In: *ICML*. 2024.

- [7] **E. P. Chandler**, S. Shoushtari, J. Liu, M. S. Asif, and U. S. Kamilov. “Overcoming Distribution Shifts in Plug-and-Play Methods with Test-Time Training”. In: *IEEE CAMSAP*. 2023.

Preprints (*Equal contribution/Shared first authorship)

- [8] S. Shoushtari, **E. P. Chandler**, X. Shi, and U. S. Kamilov. *Fast and Faithful: Principled Conditional Flow Matching for Inverse Problems*. 2026. arXiv: 2609.12953 [cs.CV]. URL: <https://arxiv.org/abs/2609.12953>.
- [9] X. Shi*, **E. P. Chandler***, C. Y. Park, S. Shoushtari, and U. S. Kamilov. *NullFlow: One-Step Generative Reconstruction*. arXiv preprint arXiv:2606.22696. 2026.
- [10] C. Y. Park, **E. P. Chandler**, Y. Hu, M. T. McCann, C. Garcia-Cardona, B. Wohlberg, and U. S. Kamilov. *Stochastic Generative Plug-and-Play Priors*. arXiv preprint arXiv:2604.03603. 2026.
- [11] S. Shoushtari, **E. P. Chandler**, Y. Wang, M. S. Asif, and U. S. Kamilov. *Unsupervised Detection of Distribution Shift in Inverse Problems using Diffusion Models*. arXiv preprint arXiv:2505.11482. 2025.
- [12] **E. P. Chandler**, S. Shoushtari, B. Wohlberg, and U. S. Kamilov. *Analysis Plug-and-Play Methods for Imaging Inverse Problems*. arXiv preprint arXiv:2509.15422. 2025.

Technical Skills

Programming & Tools: Python (PyTorch, NumPy), MATLAB, Linux/Bash, Git, Docker

Domain Knowledge: Computational Imaging, Deep Learning, Optimization, Inverse Problems, and Computer Vision